

Bagging

Bagging (Bootstrap Aggregation) reduces variance by training multiple models in parallel on random data subsets and averaging results (eg. Random Forest), ideal for overfitted data.

Bagging (Random Forest)

Method - Trains many models in parallel, independently, on random subsets of the data with replacement.

Purpose - Reduces high variance (overfitting)

Combination - Averages or takes a majority vote of all models.

Use case - When the classifier is unstable and has high variance

Core Mechanism

1) Bootstrapping - Multiple random subsets are created from the original dataset by sampling with replacement.

This means a single data point can appear multiple times in one subset or not at all (the $\sim 34\%$ excluded are called "out of bag" samples)

2) Parallel Training

A separate base model (eg. decision Tree) is trained independently and in parallel on each of these bootstrap samples.

3) Aggregation

The predictions from all individual models are combined to produce a final result

1) classification: Uses majority voting (the most frequent class wins)

2) Regression: Uses averaging (the mean of all predicted values)

Key Advantages & Disadvantages

Variance Reduction - Effectively reduces the noise & sensitivity to specific data points, leading to better generalization.

Parallelization - Since models are independent, they can be trained simultaneously, saving time on multi-

core systems.

stability - It is robust against outliers and missing values.

Interpretability - Combining many models makes it harder to explain exactly how a specific decision was reached.

Computational cost - Training hundreds of models (like 500 trees in a forest) is memory and CPU-intensive.

Common Algorithms and Variations

Random Forest - The famous bagging application, it uses decision trees and adds "feature randomness" to further

de-correlate the models.

Pasting - Similar to bagging, but sampling is done without replacement

Random Subspaces: Each model is trained on a random subset of features instead of data rows.

Random Patches - Combines both bagging (row sampling) and random subspaces (feature sampling)